TECHNOLOGY EVALUATION MATRIX

From Organizational Problem to Technology

Research Title: WEB-TRANSCO: A Web-Based Booking Platform for Jeepney Transport Services

Course Outcome: CO2 – Evaluate the suitability, feasibility, and implications of adopting emerging technologies

A. Week 4 Organizational Problem / Opportunity

Jeepney transport groups face significant inefficiencies in managing bookings for cargo, shuttle services, and private rides. Customers currently rely on manual arrangements (phone calls, word-of-mouth, or walk-ins), which result in delays, miscommunication, missed opportunities, and limited service accessibility. A digital solution is needed to streamline booking, scheduling, and payment processes while expanding services beyond traditional passenger trips.

B. Technology Options

Three emerging technologies were selected for evaluation based on their relevance to digital booking platforms, fleet operations, and the specific context of jeepney transport modernization in the Philippines:

1. Cloud Computing – Provides scalable infrastructure, centralized data storage, web hosting, and secure payment processing for a booking platform accessible anytime and anywhere.

2. Artificial Intelligence (AI) – Enables intelligent demand forecasting, automated scheduling and route optimization, dynamic pricing, chatbot support, and predictive insights for operators.

3. Internet of Things (IoT) – Supports real-time vehicle tracking, GPS-based location services, telematics for fleet status, and sensor data for operational visibility and safety.

C. Technology Evaluation Matrix

Each technology is rated on a 1–5 scale (5 = Excellent / Highly Suitable; 4 = Very Good / Suitable; 3 = Moderate / Potentially Suitable; 2 = Limited Suitability; 1 = Not Suitable). Short justifications accompany each rating.

Evaluation Criteria	Cloud Computing	Artificial Intelligence (AI)	Internet of Things (IoT)
Organizational Fit	**Rating: 5/5** Directly addresses the core need for a centralized, always-available booking system. Enables web-based reservations for private rides, cargo, and shuttles without requiring heavy local infrastructure.	**Rating: 4/5** Strongly supports smarter booking by predicting demand and optimizing schedules, reducing miscommunication and idle capacity. Complements rather than replaces the core booking need.	**Rating: 4/5** Addresses real-time visibility gaps (location, availability). Highly relevant for tracking jeepneys during bookings but is more operational than the primary booking interface itself.
Business Value	**Rating: 5/5** Creates new revenue streams through digital bookings, reduces operational friction, improves utilization, and strengthens competitiveness against informal arrangements. Supports scalable growth of shuttle and cargo services.	**Rating: 5/5** Delivers measurable value via demand prediction (higher fill rates), dynamic pricing, reduced empty trips, and automated customer support. Can increase revenue and cut costs significantly once data is available.	**Rating: 4/5** Improves fleet utilization, reduces downtime through better monitoring, and enhances customer trust via live tracking. Value is high but depends on hardware deployment across the fleet.
Benefits	**Rating: 5/5** Centralized booking and payments; anytime access for customers and operators; reduced miscommunication; easier expansion of services; lower reliance on paper/manual processes; foundation for future digital features.	**Rating: 4/5** Optimized routes and schedules; predictive demand insights; 24/7 chatbot assistance; personalized recommendations; reduced administrative workload; potential for fewer delays and higher customer satisfaction.	**Rating: 4/5** Real-time vehicle location and status; improved safety and accountability; better coordination of private/cargo trips; data for maintenance and route planning; enhanced transparency for customers.
Feasibility	**Rating: 5/5** Highly feasible. Web applications hosted on public cloud platforms (AWS, Azure, Google Cloud, or local providers) are mature, widely available, and can start with modest resources. Many transport	**Rating: 3/5** Moderately feasible. Requires quality historical and real-time data, skilled personnel or external partners for model development, and ongoing refinement.	**Rating: 3/5** Moderately feasible. Requires installation of GPS/telematics devices on vehicles, reliable connectivity, and integration with a backend platform. Upfront hardware costs and

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	cooperatives already use basic digital tools.	Full value may take time to realize for smaller jeepney groups.	maintenance can be barriers for small operators.
Cost / Resources	**Rating: 4/5** Moderate and scalable. Pay-as-you-go cloud pricing allows starting small. Requires web development skills, domain, basic security configuration, and ongoing hosting fees. Lower barrier than building on-premise systems.	**Rating: 2/5** Higher resource demand. Needs data scientists/ML engineers (or vendor solutions), training data, compute resources for model training, and continuous monitoring. Cost can rise with sophistication.	**Rating: 2/5** Significant initial investment in GPS trackers, sensors, SIM cards, and installation. Ongoing data plans and device maintenance add recurring costs. May require subsidies or phased rollout for jeepney cooperatives.
Security and Privacy	**Rating: 4/5** Cloud providers offer strong built-in security (encryption, IAM, compliance certifications). Main concerns: proper configuration, secure payment gateways, protection of customer booking data, and access controls for operators.	**Rating: 3/5** AI systems process sensitive trip and user data. Risks include model inversion, bias in recommendations, and potential misuse of predictive insights. Requires careful data governance and anonymization practices.	**Rating: 3/5** Continuous location tracking raises privacy concerns for drivers and passengers. Device vulnerabilities and data interception are risks. Strong encryption, consent mechanisms, and secure device management are essential.
Ethical Implications	**Rating: 4/5** Generally positive if access is inclusive (mobile-friendly, low-bandwidth options). Risk of excluding operators or customers with limited digital literacy or connectivity. Fair pricing and transparent terms are important.	**Rating: 3/5** Potential for algorithmic bias in scheduling or pricing that could disadvantage certain routes, drivers, or customer groups. Transparency and human oversight of automated decisions are ethically necessary.	**Rating: 3/5** Continuous surveillance of vehicle movements can feel intrusive. Drivers may resist monitoring. Clear policies on data use, retention, and purpose limitation are required to maintain trust.
Governance	**Rating: 4/5** Needs clear data ownership policies, user access roles, service-level agreements with cloud providers, backup/recovery plans, and compliance with local data protection regulations (e.g., Philippine Data Privacy Act).	**Rating: 3/5** Requires policies on model accountability, audit trails for automated decisions, regular bias testing, and guidelines for when human intervention overrides AI recommendations.	**Rating: 3/5** Requires device inventory management, security update policies, data retention rules for location history, and agreements with drivers/cooperatives on monitoring practices.
Sustainability	**Rating: 4/5** Supports organizational sustainability by reducing paper use and enabling better resource planning. Cloud data centers have environmental footprints, but efficient digital operations can lower overall fuel waste from empty or poorly coordinated trips.	**Rating: 4/5** Route and load optimization can reduce fuel consumption and emissions. Predictive maintenance (if combined with data) extends vehicle life. Positive contribution to environmental and economic sustainability of the fleet.	**Rating: 4/5** Real-time monitoring and efficient dispatching reduce unnecessary travel and idling. Supports safer, more reliable operations that encourage modal shift toward organized public transport, aligning with sustainable mobility goals.
Risks and Limitations	**Rating: 4/5** Dependence on internet connectivity; vendor lock-in risk; need for digital skills among operators; potential resistance to change from traditional booking habits. Mitigable with offline-capable features and training.	**Rating: 3/5** Data scarcity in early stages limits model accuracy; black-box decision-making; high skill requirements; possible over-reliance on automation. Performance depends on continuous data quality and domain expertise.	**Rating: 2/5** Hardware failure or theft; connectivity gaps in some areas; battery/power issues; higher total cost of ownership; integration complexity with existing vehicles. Adoption may be slow without incentives.
Overall Suitability	**Rating: 5/5** Most suitable foundational technology. Directly solves the booking inefficiency problem, is realistic for jeepney groups to adopt, offers strong business value, and provides the platform on which AI and IoT features can later be added.	**Rating: 3/5** Highly valuable as an enhancement layer once a reliable booking and data platform exists. Less suitable as the sole or primary solution for the immediate organizational problem of manual booking processes.	**Rating: 3/5** Excellent complementary technology for operational visibility and customer experience. Not the primary solution for creating a booking interface and payment capability, which is the core need identified.

Summary of Overall Ratings: Cloud Computing – 5 | Artificial Intelligence – 3 | Internet of Things – 3

Comparison of Technology Options

Cloud Computing demonstrates the strongest organizational fit because it directly enables the centralized web-based booking platform required to replace manual arrangements. It also provides the greatest immediate business value through improved accessibility, reduced miscommunication, and new service channels (cargo, private rides, shuttles). Feasibility is highest for Cloud Computing due to mature, pay-as-you-go services and relatively lower skill and capital barriers compared with full AI model development or fleet-wide IoT hardware deployment.

AI offers the highest potential long-term efficiency gains (demand prediction, optimization) but depends on having a functioning data platform first. IoT delivers strong operational visibility and supports

sustainability through better fleet utilization, yet its higher upfront costs and privacy considerations make it less ideal as the initial primary investment. Ethical and security concerns are manageable for all three technologies when proper governance, encryption, consent, and compliance with the Philippine Data Privacy Act are implemented.

D. Final Technology Recommendation

The most appropriate technology for addressing the identified organizational problem is Cloud Computing. It is highly suitable because it provides the scalable, always-available infrastructure needed to host a centralized web-based booking platform for private rides, cargo transport, and shuttle services—directly solving the inefficiencies of manual arrangements. Expected business value includes increased booking volume, reduced operational delays and miscommunication, expanded service offerings, and a foundation for future digital features that strengthen competitiveness and create new revenue streams for jeepney transport groups. An important consideration is the need for strong security configuration, data privacy compliance, digital literacy training for operators, and reliable internet access to ensure inclusive and sustainable adoption.

E. Sources

1. *Mnyakin, M. (2023). Applications of AI, IoT, and Cloud Computing in smart transportation: A review.* Artificial Intelligence in Society. https://www.researchberg.com/index.php/ai/article/view/108
2. Saigon Technology. (2026). IoT in transportation and logistics: 12 use cases, benefits & ROI guide. https://saigontechnology.com/blog/iot-in-transportation-and-logistics/
3. *Burlat, M. P. (2026). JeepniGo: A smart mobility platform for accessible, inclusive, and sustainable jeepney commuting in the Philippines.* SSRN. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=6620819
4. Developers.dev. (2025). Building the taxi apps of tomorrow: AI, IoT, and strategic roadmaps. https://www.developers.dev/tech-talk/building-the-taxi-apps-of-tomorrow-ai-iot-and-beyond.html
5. Course materials and concepts from Cengage MindTap / assigned textbook chapters on emerging technologies, cloud computing, AI applications, and technology evaluation frameworks (organizational fit, feasibility, ethics, security, governance, and sustainability).

Note: *This evaluation builds directly on the Week 4 Research Topic Proposal for WEB-TRANSCO. Ratings and the final recommendation reflect critical analysis of suitability for jeepney transport groups, balancing immediate organizational needs with longer-term technological opportunities.*